PS-029

Application-Oriented Network Quality Evaluation Using QoO

Team DevElite

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Problem Description

Raw QoS numbers do not tell us whether an application will work well.

- Bandwidth and round-trip time describe the network, not what a given application can tolerate.
- The same path can suit bulk transfer yet fail real-time video.
- QoO scores a network against explicit application requirements (profiles).
- How much the choice of profile changes the verdict is rarely measured.
- Ad-hoc impairment testing makes cross-profile comparison unreliable.

Background

From QoS to QoE to QoO

QoS	QoE	QoO
Measures the network: bandwidth, delay, jitter, loss.	Measures user perception. Needs user studies; out of scope here.	Scores network quality against an application's stated needs using IETF-defined profiles.

This project stays within the brief: no new QoO mathematics, no universal quality score, and no user-experience claims without user testing.

Literature & RFC Review

Existing standards and tools this work builds on.

- IETF Quality of Outcome (QoO) Internet-Draft: defines application profiles and the scoring approach [1].
- IETF 126 hackathon QoO testbed: container lab with a gateway, collector, InfluxDB and Grafana [2].
- IPPM metrics for delay, jitter and loss: RFC 2681, RFC 3393, RFC 7680 [3].
- RFC 6349 on TCP throughput testing, for the throughput measurements [4].
- goDASH adaptive streaming client for application-level metrics [5].

Proposed Solution

Drive the supplied QoO testbed with a repeatable experiment harness.

1

Two profiles

Mentor-approved; every parameter recorded.

2

Six scenarios

Delay, jitter and loss applied identically to both.

3

Measure

QoO, latency, throughput and one application metric.

4

Analyse

Dashboard and report on how profile choice changes the verdict.

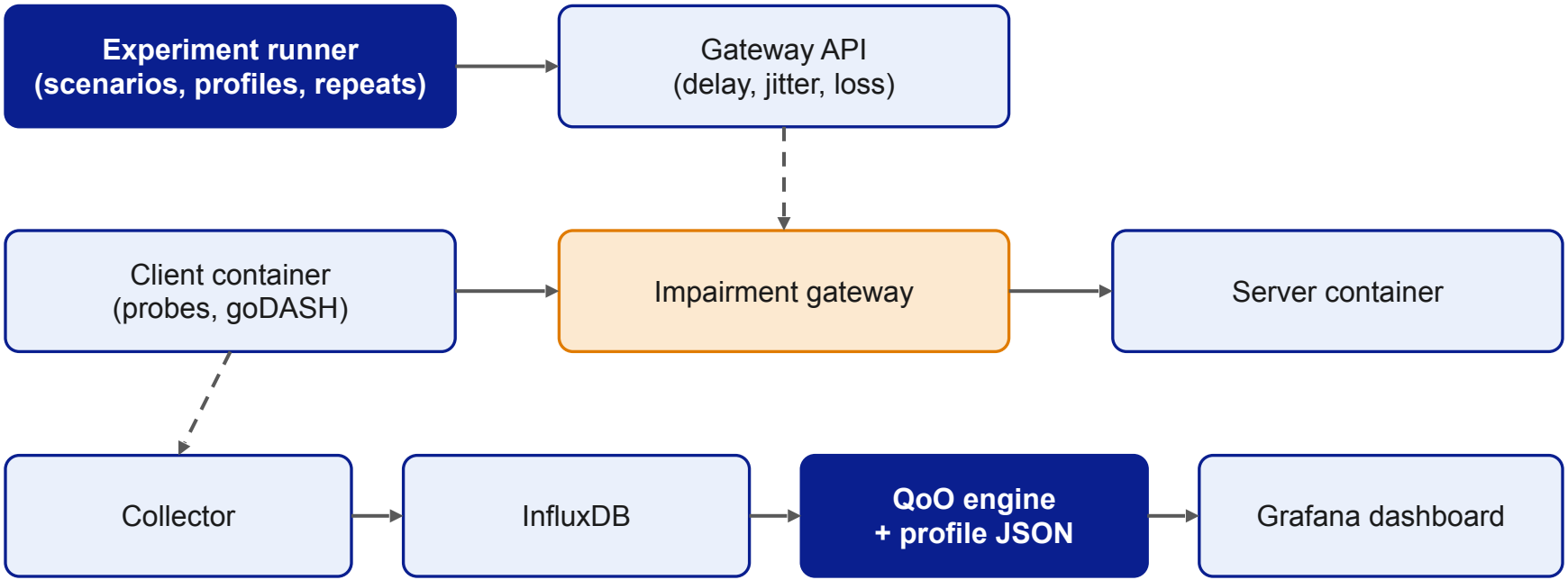
Outputs: harness, dataset with parameter records, results dashboard, comparative report and demo.

Optimization Proposed

Make the experiments scripted, repeatable and comparable.

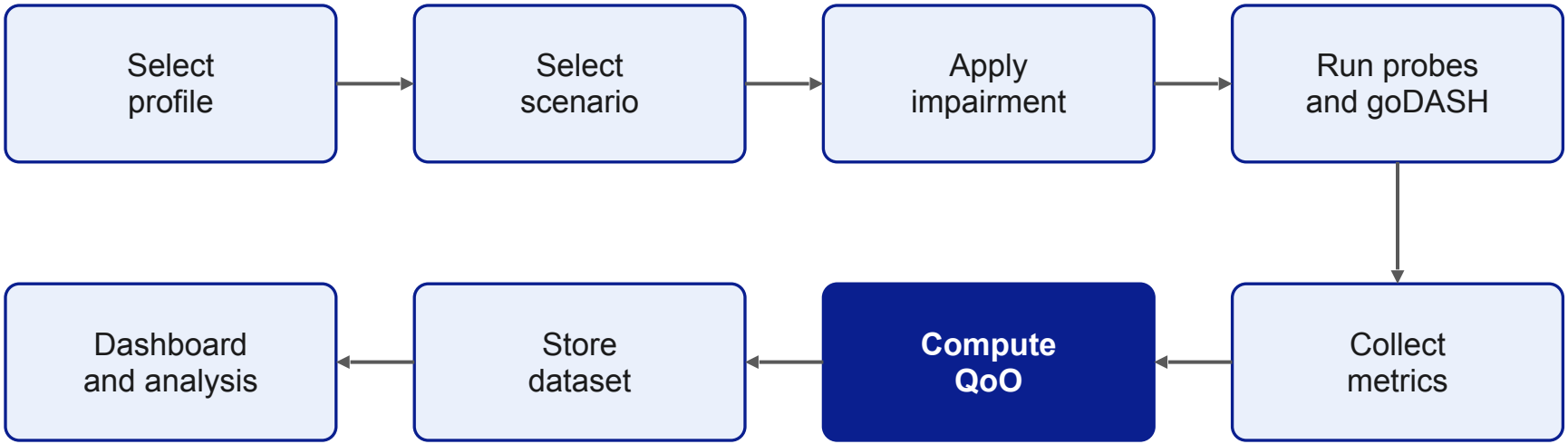
Aspect	Before	After
Scenario setup	Manual commands	Scripted, parameterised scenarios
Repeatability	Single ad-hoc runs	N repeats with mean and variance
Parameter tracking	Scattered notes	Logged per experiment (CSV and JSON)
Comparison	Manual spreadsheet work	Dashboard, profiles side by side

Solution Architecture & Design



Dashed arrows: control and measurement feeds. Solid arrows: traffic and data flow.

Data Flow Diagram



Repeat for every run, scenario and profile; all parameters are logged with each row.

Deployment Timeline

M1	**11 – 14 Oct**	Level 1 submission: repo, PDF, video, transfer
M2	**12 – 15 Oct**	Testbed running, profiles shortlisted
M3	**16 – 18 Oct**	Profiles approved, six scenarios, data schema
M4	**16 – 20 Oct**	Experiment runner works for one profile
M5	**21 – 25 Oct**	Full runs, dashboard, presentation update
M6	**26 – 28 Oct**	Analysis, demo link, faculty sign-off
M7	**29 – 30 Oct**	Final submission check and buffer

AIORI dates: Steps 1-7 due 14 Oct; mentoring and Steps 8-10 from 16 to 31 Oct; evaluators from 31 Oct.

Bibliography

[1] IETF IPPM Working Group, “Quality of Outcome (QoO)”, Internet-Draft (draft-ietf-ippm-qoo; check current version).

[2] IETF 126 Hackathon QoO testbed, github.com/cj-ike-kunze/ietf-126-hackathon-qoo.

[3] RFC 2681 (round-trip delay), RFC 3393 (delay variation), RFC 7680 (one-way loss), IETF.

[4] RFC 6349, Framework for TCP Throughput Testing, IETF.

[5] goDASH, DASH streaming client, github.com/uccmisl/godash.

[6] Linux tc-netem, network emulation documentation.



Thank You for Your Attention!

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